

MedFusion: A Unified Vision-Language Framework for Radiology Images

PHASE II REPORT

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BONAFIDE CERTIFICATE

Certified that this report titled “**MedFusion: A Unified Vision-Language Framework for Radiology Images**” is the bonafide work of **Akhshan P** (Reg. No: **21011101014**), **Deepak T** (Reg. No: **21011101039**), **Janani Harshatha J** (Reg. No: **21011101053**), **Karthick N G** (Reg. No: **21011101058**) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Recent advancements in artificial intelligence (AI) have unlocked transformative potential in healthcare, particularly in automating diagnostic processes through medical imaging. Despite this, current systems face critical limitations in adaptability across imaging modalities, computational efficiency, and domain-specific data availability. In this study, we introduce a novel AI-powered system designed to address these challenges and enhance diagnostic precision through tasks including medical visual question answering (VQA), medical report generation, and disease identification. Our model leverages a multi-encoder architecture.

Our embedding strategy concatenates outputs from three specialized networks: Pyramid V2 Transformer trained using CLIP contrastive learning on ROCO dataset and custom medical images, ConvNeXt fine-tuned on medical images with its final layer removed for feature extraction, and CLIP (ViT-32B) fine-tuned on MedICaT-200k and ROCO-80k. This combined embedding is then passed to Med-LLaMA, our adaptation of LLaMA 3 fine-tuned on PubMedQA, then complete multimodal system is trained on PMC VQA dataset, achieving great generalization.

Our empirical evaluations reveal substantial performance gains, particularly in generating precise, contextually relevant medical reports and accurately answering domain-specific questions, achieving state-of-the-art results on VQA and report generation benchmarks. This model significantly advances the field of automated medical image interpretation, providing clinicians with an advanced, adaptable tool that offers immediate diagnostic insights, thereby improving clinical workflows and patient outcomes across diverse medical settings.

Keywords - Medical Image Analysis, Artificial Intelligence, Deep Learning, Pyramid V2 Transformer, ConvNeXt, CLIP, LLaMA 3, Med-LLaMA

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LIST OF SYMBOLS, ABBREVIATIONS

τ Temperature parameter

CLIP Contrastive Language Image Pre-training

CNN Convolution Neural Network

CT Computed Tomography

EVA Exploring Visual Architecture

LLAMA Large Language Model Meta AI

Med-LLaMA Medical Large Language Model Meta AI

MedICaT Medical Image-Caption-Tabular dataset

MRI Magnetic Resonance Imaging

NLP Natural Language Processing

PMC VQA PubMed Central Visual Question Answering

PubMedQA PubMed Question Answering

ROCO Radiology Objects in Context

SLAKE Semantic Language Annotations for Knowledge Extraction

ViT Vision Transformer

VQA Visual Question Answering

CHAPTER 1: INTRODUCTION

1.1 Overview

Medical imaging technology has revolutionized healthcare by enabling non-invasive visualization of internal body structures for diagnosis and treatment planning. However, the increasing volume of medical images, coupled with a shortage of qualified radiologists, has created significant challenges in healthcare delivery. Our model addresses these challenges by leveraging advanced artificial intelligence techniques to automate the analysis of medical images and generate comprehensive medical reports.

Our system combines state-of-the-art vision and language models to process various types of medical images, including X-rays, MRIs, and CT scans. By automating the analysis process, our model aims to reduce the time between image acquisition and diagnosis, potentially improving patient outcomes through earlier intervention and treatment.

1.2 Problem Formulation

The current healthcare landscape faces several critical challenges in medical image analysis:

- **Resource Constraints:** The growing demand for medical imaging has outpaced the availability of qualified radiologists, leading to significant delays in diagnosis and treatment.
- **Accessibility Gap:** Rural and underserved areas often lack access to specialized medical professionals, resulting in delayed or incomplete diagnoses.
- **Consistency and Accuracy:** Human interpretation of medical images can be subject

to variability and fatigue-induced errors, potentially affecting diagnostic accuracy.

- **Time Efficiency:** Traditional workflows require substantial time for image analysis and report generation, which can delay critical treatment decisions.
- **Data Complexity and Volume:** Medical images are complex and data-intensive. Advanced computational techniques are required to process and analyze these large datasets efficiently.
- **Integration of Multimodal Data:** Combining data from multiple modalities (e.g., radiology, pathology) is technically challenging but essential for a comprehensive understanding of a patient's condition.
- **Standardization and Interoperability:** Lack of standardization in medical image formats and data annotation hinders effective use across healthcare systems. Establishing standardized protocols is crucial for seamless data exchange.
- **Data Privacy and Security:** Medical images and patient data are highly sensitive. Robust data protection measures and compliance with regulatory requirements are essential to safeguard patient information.

1.3 Objectives

The primary objectives of this project are:

- To develop an AI-powered system capable of analyzing various types of medical images and generating accurate diagnostic reports.
- To implement and optimize state-of-the-art vision and language models (Pyramid-V2 Transformer trained using CLIP and MedLLAMA) for medical image analysis.
- To create a system that can process images and generate reports in multiple languages, improving accessibility in diverse healthcare settings.

1.4 Motivation

The motivation for developing our model stems from several critical factors in contemporary healthcare:

- **Healthcare Access:** In rural and underserved areas, the lack of specialized medical professionals creates significant barriers to timely diagnosis. Our Model aims to bridge this gap by providing rapid, AI-assisted image analysis capabilities.
- **Efficiency Improvement:** By automating the initial analysis of medical images, the system can significantly reduce the time between image acquisition and preliminary diagnosis, enabling healthcare providers to make more timely decisions.
- **Support for Medical Professionals:** Rather than replacing human expertise, our Model serves as a powerful tool to assist medical professionals, helping them work more efficiently and focus their attention on complex cases requiring human judgment.
- **Technological Innovation:** The integration of advanced AI technologies in healthcare represents a significant step forward in medical diagnostics, potentially leading to improved patient outcomes and more efficient healthcare delivery systems.

CHAPTER 2: LITERATURE SURVEY

2.1 Medical Image Analysis in Healthcare

Medical image analysis is a critical component of modern healthcare diagnostics, enabling healthcare professionals to visualize and interpret internal body structures for accurate diagnosis and treatment planning. Traditional analysis methods rely heavily on human expertise, which can be time-consuming and subject to variability. The integration of artificial intelligence in medical image analysis represents a significant advancement in healthcare technology, offering potential solutions to these challenges.

2.2 Related Works

Medical Visual Question Answering (MedVQA) seeks to automate image interpretation for clinical applications. Traditional classification-based methods efficiently address closed-ended questions but often struggle with the complexity of open-ended queries. To overcome this, recent approaches employ generative frameworks, treating MedVQA as a text generation task to enhance flexibility. Notably, the MISS framework combines multi-task learning with the TransCap method to expand available training data through large language models (LLMs), improving performance even with limited multimodal datasets .

Hybrid approaches, such as Q2ATransformer, bridge classification and generation by leveraging Transformer-based architectures with learnable answer embeddings, yielding improved accuracy on both open- and closed-ended questions. Additionally, attention-based fusion strategies enhance MedVQA models' ability to process complex medical images, while large datasets like PMC-VQA offer new benchmarks for pretraining and evaluation . These innovations underscore the importance of multimodal feature alignment and hybrid

modeling to advance MedVQA’s applicability in clinical settings.

Recent developments in vision transformers have significantly enhanced feature extraction capabilities for medical images. The Pyramid Vision Transformer V2 architecture has gained attention for its innovative approach to handling medical imagery through overlapping patches. Unlike conventional vision transformers that process non-overlapping image patches, the overlapping patch strategy in Pyramid V2 enhances the model’s ability to capture fine-grained details and contextual relationships between adjacent regions in medical images, which is particularly crucial for identifying subtle pathological changes. ConvNeXt architectures, fine-tuned specifically on medical imaging datasets, have demonstrated exceptional performance in extracting unique visual features from complex medical scans. By removing the final classification layer and utilizing these networks as feature extractors, researchers have developed powerful embedding generators tailored to the nuances of medical imagery. This adaptation allows for more effective representation of anatomical structures and pathological indicators than general-purpose vision models.

The integration of CLIP (Contrastive Language-Image Pre-training) methods has marked a significant advancement in multimodal medical understanding. CLIP-based models fine-tuned on specialized medical datasets such as MedICaT-200k and ROCO-80k have demonstrated remarkable capability in aligning visual features with medical terminology. This alignment facilitates more accurate interpretation of medical images in the context of clinical queries, bridging the semantic gap between visual data and medical knowledge . The integration of advanced visual embedding techniques with specialized language models creates powerful multimodal architectures for medical visual question answering. Training these integrated systems on comprehensive datasets like PMC-VQA enables the model to learn complex relationships between visual medical data and clinical queries. This approach addresses the fundamental challenges in MedVQA: the need for precise visual feature extraction, accurate medical knowledge representation, and effective alignment between visual and textual modalities.

CHAPTER 3: METHODOLOGY

3.1 Base Architecture Overview

MiniGPT-Med[1] is a multimodal model designed to assist with radiological diagnosis by bridging the gap between medical imaging and natural language understanding. Built upon the MiniGPT-v2 framework, it combines a frozen vision encoder with a large language model (LLaMA2-Chat) to support a wide range of tasks such as medical report generation, disease detection, and visual question answering (VQA) as given in Fig.3.1. By integrating task-specific instructions and visual inputs, MiniGPT-Med can handle both grounding (spatial localization) and non-grounding (descriptive or reasoning-based) tasks within a single unified system. This flexibility allows it to act as a general-purpose interface for radiology diagnostics, significantly enhancing the efficiency and accuracy of interpreting medical images.

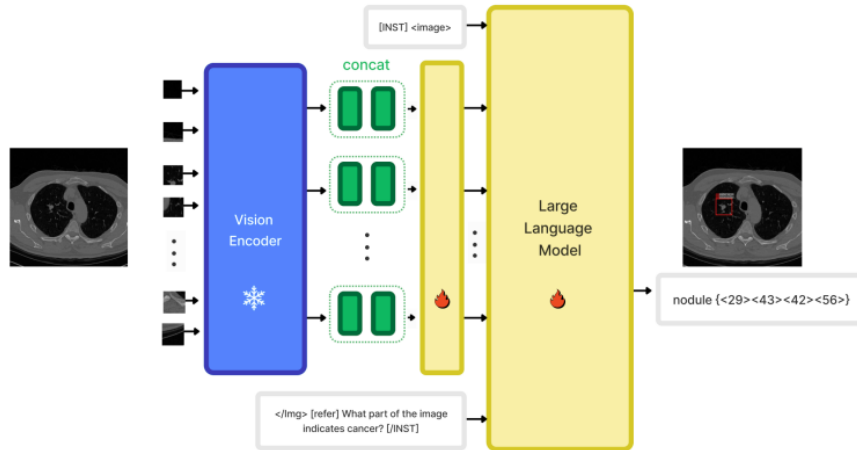


Figure 3.1: Base Architecture

3.1.1 Visual Encoder (EVA)

The visual encoder uses EVA, a vision transformer trained using CLIP (Contrastive Language image pretraining)[2], as its visual backbone. This encoder is pre-trained and remains frozen during training to preserve its strong feature extraction capabilities. It processes high-resolution medical images (448×448) and outputs rich visual representations. To adapt to the image size, the positional encodings are interpolated, ensuring compatibility with the input dimensions.

3.1.2 Token Merging and Linear Projection

To improve computational efficiency and manage the large volume of visual data, the four adjacent visual tokens are merged into one. This reduces the sequence length while preserving spatial information. These merged visual tokens are then passed through a linear projection layer, which maps them into the token space of the language model, allowing seamless integration with text-based prompts.

3.1.3 Large Language Model (LLaMA2-Chat 7B)

The core of the model’s language understanding and generation lies in LLaMA2-Chat (7B) [8], which is fine-tuned using Low-Rank Adaptation (LoRA). This open-source LLM has been trained on extensive medical and general textual data and serves as the model’s reasoning engine. It interprets prompts, generates reports, answers questions, and identifies abnormalities within the image context.

3.2 Architecture Overview

The implemented model utilizes a multi-encoder visual system combined with Med-LLaMA for language processing, creating a system capable of accurate, context-aware responses to medical questions. By employing the Pyramid V2 Transformer with overlapping patches, ConvNeXt, and fine-tuned CLIP (ViT-32B), the model effectively captures com-

prehensive visual representations and marginalizes the differences in images in the vector space. The feature concatenation layer further optimizes mapping between visual tokens and textual data, ensuring consistent representation across modalities as shown in Fig.3.2. Med-LLaMA’s domain-specific language generation then provides precise, nuanced answers by interpreting complex medical terminology. Together, these components form a high-performing, adaptable MedVQA system suited to the varied demands of clinical applications.

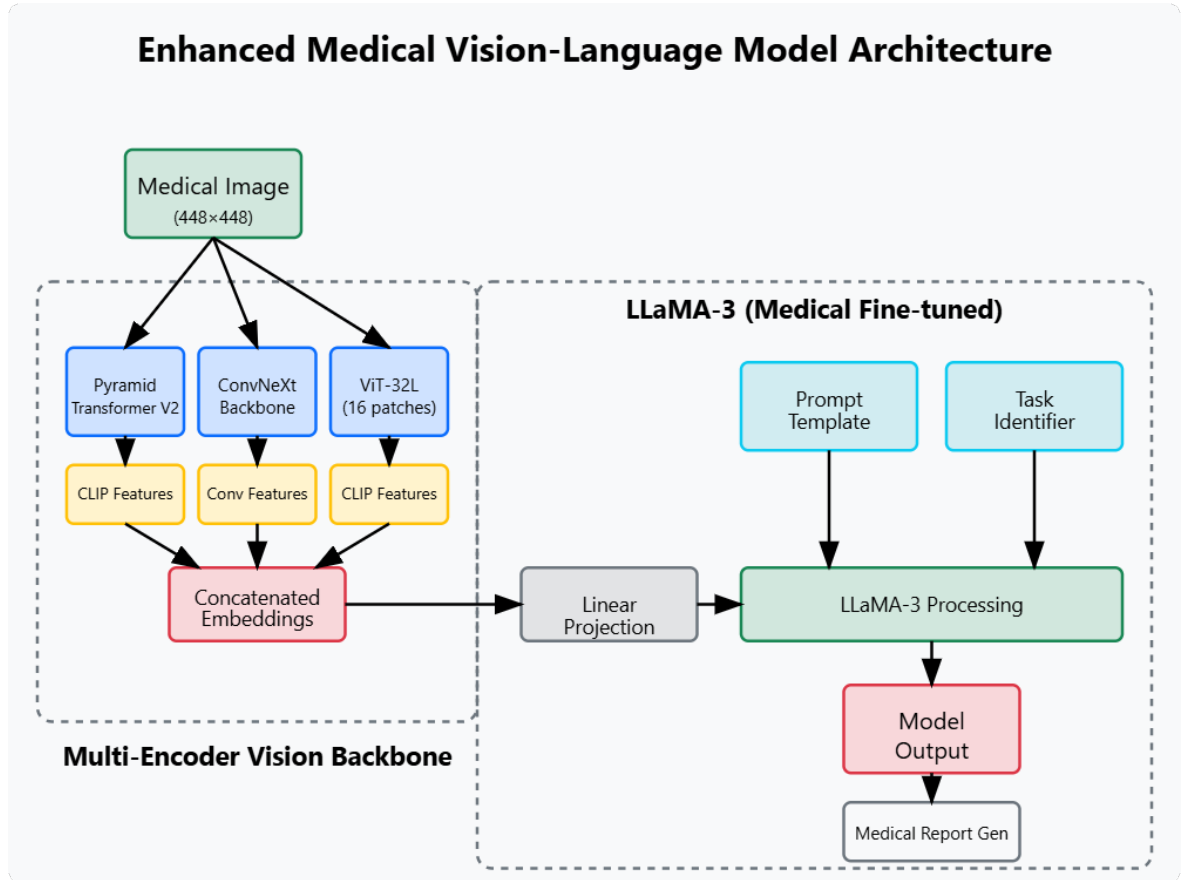


Figure 3.2: Enhanced MedVQA Architecture

3.2.1 Pyramid V2 Transformer Implementation

The Pyramid V2 Transformer [15] is a vision model that builds on traditional transformer architectures by incorporating overlapping patches to capture fine-grained visual patterns and difference-based features. This approach enables the model to detect subtle differences in medical images that might indicate pathological conditions. During the training process,

the Pyramid V2 Transformer is trained using CLIP contrastive learning methodology on the ROCO dataset and custom medical images, learning to associate images with their corresponding textual descriptions by maximizing the similarity between matching image-text pairs while minimizing the similarity between non-matching pairs.

One of the key advantages of the Pyramid V2 Transformer is its ability to process overlapping patches, which allows it to maintain spatial continuity and capture edge information more effectively than traditional non-overlapping patch transformers. This is particularly valuable for medical imaging where boundary detection and structural continuity are crucial for accurate diagnosis. The overlapping patch mechanism helps establish relationships between adjacent regions in medical images, providing a more coherent representation of anatomical structures and pathological findings. During implementation the architecture given in Fig.3.3, it is configured it with:

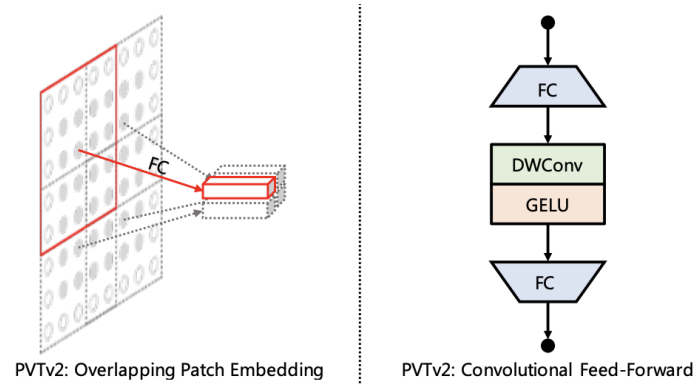


Figure 3.3: Pyramid V2 overlapping patches

- 16×16 pixel patches with 8-pixel overlap to maintain spatial continuity
- 12 transformer layers with 12 attention heads each
- Hidden dimension of 768 for rich feature representation
- Layer-wise feature pyramid to capture multi-scale information

3.2.2 ConvNeXt (Local Feature Extractor)

ConvNeXt[16] is a CNN-based encoder that uses depthwise separable convolutions and inverted bottleneck layers to process images at native resolution. By preserving fine details, ConvNeXt outputs high-resolution feature maps that provide pixel-level precision for spatial tasks like tumor segmentation. Unlike models that lack dedicated spatial modeling, ConvNeXt's design enables it to capture complex patterns and features in medical images. Its unique feature, dynamic kernel scaling, adjusts receptive fields based on input content, allowing it to effectively handle varying image sizes and resolutions. The ConvNeXt architecture given in Fig.3.4 configuration included:

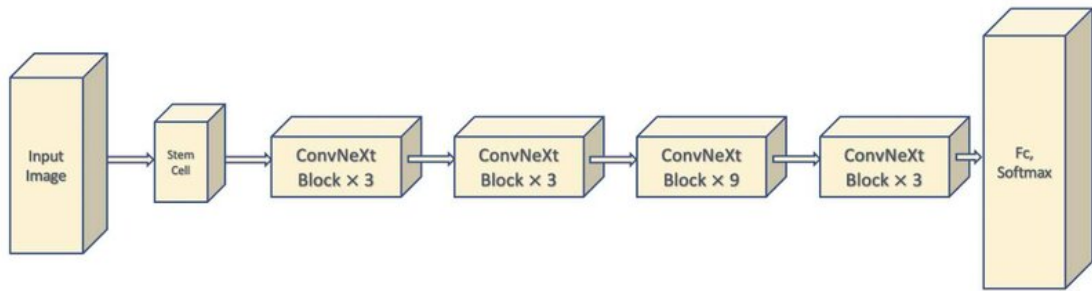


Figure 3.4: ConvNeXt Architecture

- Depthwise separable convolutions with large kernels (7×7)
- Inverted bottleneck structure with expansion factor 4
- Layer normalization instead of batch normalization for stable training
- Stochastic depth rate of 0.3 for regularization

3.2.3 Vision Transformer

Vision Transformer (ViT) is a neural network architecture that applies transformer models to image classification tasks. It processes images as a sequence of patches, rather than using convolutional neural networks (CNNs) to process images as a grid of pixels. ViT uses

self-attention mechanisms to capture long-range dependencies between patches, allowing it to effectively integrate information from different parts of the image. This architecture is given in Fig.3.5.

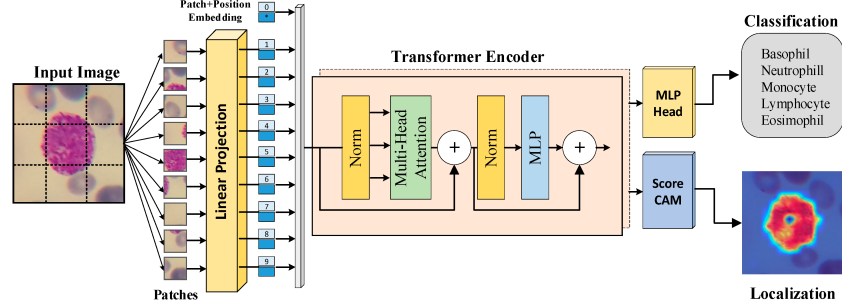


Figure 3.5: ViT Architecture

The ViT-L/16 is a 32-layer Vision Transformer that uses 16×16 patch embeddings and learnable position embeddings to handle variable image sizes. Pretrained with medical image-text pairs using CLIP loss, ViT-L/16 excels at whole-image interpretation, linking visual features to semantic meaning. Its unique feature, cross-modal attention between image patches and report keywords, allows it to effectively integrate visual and textual information

3.2.4 Feature Concatenation Layer

In our model, the feature concatenation layer[24] processes and combines the visual embeddings extracted by the three parallel encoders: Pyramid V2 Transformer, ConvNeXt, and fine-tuned CLIP (ViT-32B). Instead of using a simple linear projection, this implementation concatenates the outputs from all three networks followed by a deeper alignment network. This deeper alignment network enhances the mapping process, allowing for more complex transformations of the combined visual features.

The concatenation approach leverages the complementary strengths of each encoder: difference-based features from the Pyramid V2 Transformer, unique structural patterns from ConvNeXt, and contextual understanding from CLIP. By aligning and combining these diverse visual representations, the model ensures a comprehensive understanding of medical imagery that captures both local details and global context, which is essential for optimizing

performance across both visual and textual tasks.

3.2.5 Language Model

Med-LLaMA, an adaptation of LLaMA 3 fine-tuned on PubMedQA, generates coherent, contextually relevant medical language and provides strong performance on both generation and comprehension tasks. The block of llama is mentioned in Fig.3.6. For MedVQA applications, Med-LLaMA’s advanced architecture allows it to interpret complex medical terminology and nuanced language, making it well-suited for understanding clinical questions and generating precise answers. A key strength of Med-LLaMA in the context of multimodal applications lies in its fine-tuned ability to handle specialized medical vocabulary, allowing it to align effectively with visual features from medical images.

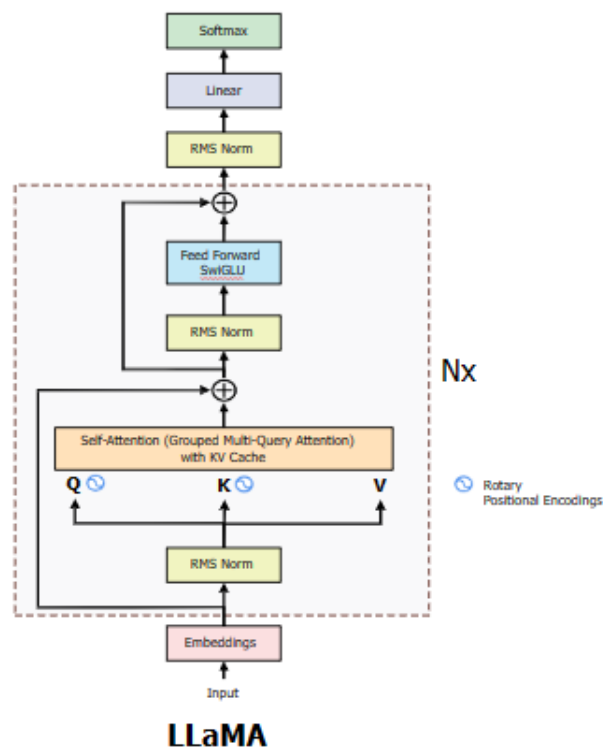


Figure 3.6: LLAMA3 Architecture

The model’s language generation capabilities are complemented by its transformer-based architecture, which efficiently manages large-scale text input, enabling it to respond accu-

rately to questions about diverse medical scenarios. Additionally, Med-LLaMA incorporates training optimizations from the LLaMA 3 foundation, which enhance performance and reduce computational costs. These improvements make Med-LLaMA not only adaptable for medical VQA tasks but also highly capable of handling the vast variety of questions in medical settings, from factual responses to more interpretive, diagnostic insights. The complete multimodal architecture is trained on the PMC VQA dataset specifically for medical visual question answering.

3.2.6 Future Enhancement

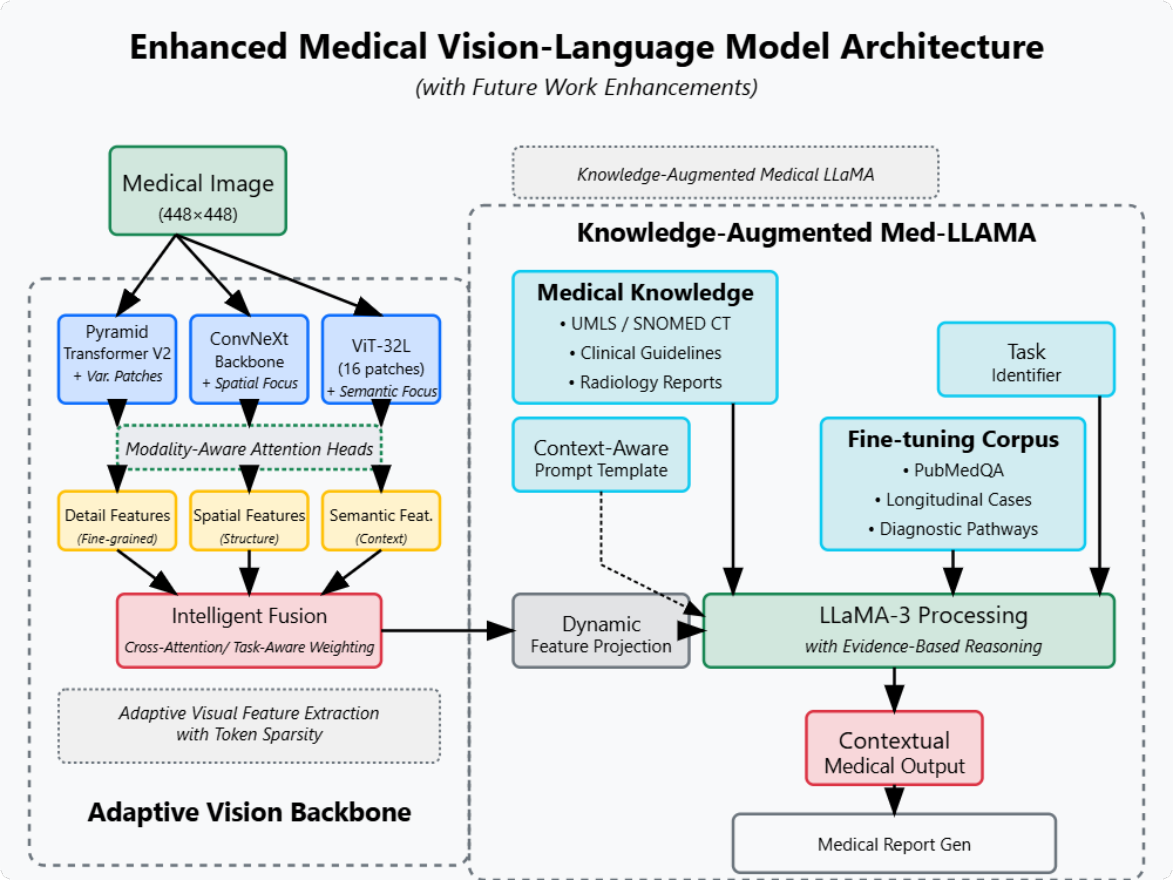


Figure 3.7: Future Works

CHAPTER 4: EXPERIMENTS

4.1 Datasets Used

4.1.1 SLAKE Dataset

The SLAKE (Structured Language-Augmented Knowledge Extraction)[4] dataset provides a foundation for medical visual question answering (MedVQA) tasks. This English-Chinese bilingual dataset consists of 642 images and 14,028 question-answer pairs, with structured annotations that include semantic labels and relationships between elements in the images.

Table 4.1: Task Types and Descriptions

Task Type	Identifier	Description/Example
Caption	[caption]	Could you describe the contents of this image for me?
VQA	[vqa]	What plane is the image in?
Detection	[detection]	pneumonia
Refer	[refer]	the nodule in the left lung
Grounding	[grounding]	describe this image in detail
Identify	[identify]	what is this 56168458

4.1.2 ROCO Dataset

The Radiology Objects in COntext (ROCO) dataset[11] is a large-scale collection of medical images from published literature, comprising over 81,000 radiology images with corresponding captions. We utilized this dataset for pretraining our image encoders, particu-

larly the Pyramid V2 Transformer. The diverse range of imaging modalities in ROCO—including X-rays, CT scans, MRIs, ultrasounds, and various specialized medical imaging techniques—makes it ideal for developing robust visual representations. The dataset’s captions contain rich clinical descriptions, enhancing the model’s ability to associate visual features with appropriate medical terminology.

4.1.3 MedICaT-200k Dataset

The MedICaT (Medical Image-Caption-Tabular dataset)[12] comprises 200,000 medical images with associated captions and tabular data extracted from scientific literature. This dataset is particularly valuable for training vision-language models as it contains rich contextual information beyond simple image-text pairs, including table references and detailed medical annotations. We leveraged MedICaT-200k for fine-tuning our CLIP (ViT-32B) component, which enabled our system to develop sophisticated understanding of medical imagery in relation to detailed clinical contexts. The dataset’s extensive coverage of specialized medical domains contributed significantly to our model’s capacity to recognize and interpret complex pathological patterns.

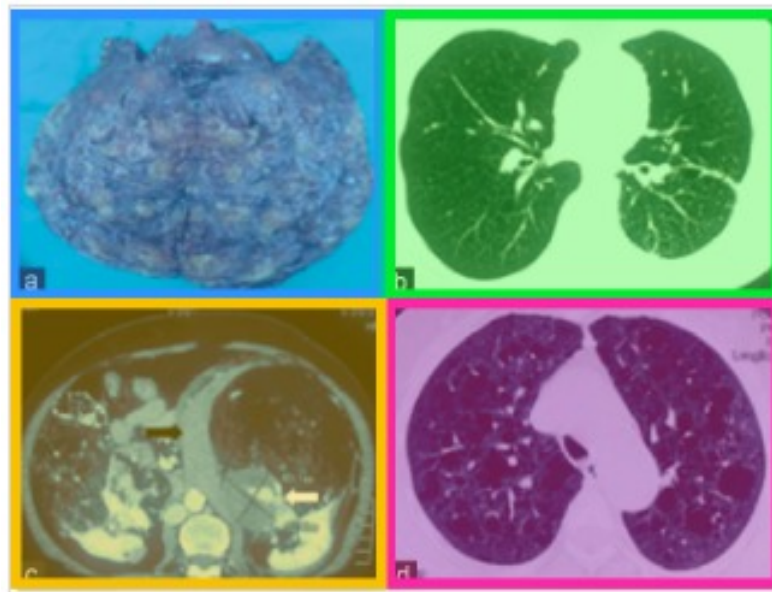
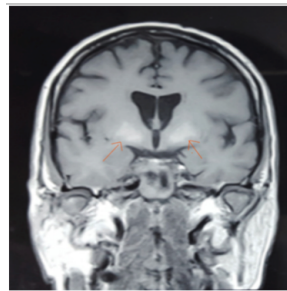


Figure 4.1: MedICAT Dataset

4.1.4 PMC VQA Dataset

The PMC VQA (PubMed Central Visual Question Answering)[13] dataset consists of medical images paired with question-answer sets derived from PubMed Central articles. This dataset specifically addresses the medical visual question answering task by providing examples of clinically relevant questions about medical images alongside expert-provided answers. We utilized PMC VQA to train our complete multimodal architecture, focusing on developing the model’s capability to answer specific diagnostic and interpretive questions about medical images. This training enabled our system to demonstrate superior performance in generating accurate and contextually appropriate responses to clinical queries about radiological imaging.



Question: What is the view of the brain used in the image?

A: Axial

B: Coronal

C: Sagittal

D: Oblique

Figure 4.2: PMC VQA Dataset

4.1.5 PubMedQA Dataset

PubMedQA is a biomedical question answering dataset [14] containing 273,000 question-answer pairs extracted from PubMed abstracts. Unlike image-based datasets, PubMedQA focuses on developing natural language understanding in the medical domain through complex biomedical questions and their corresponding evidence-based answers. We used this dataset to fine-tune our LLaMA 3 language model, creating the specialized Med-LLaMA component of our system. This fine-tuning process enhanced our model’s medical vocabulary, reasoning capabilities, and understanding of biomedical concepts, enabling more accu-

rate and contextually appropriate responses to medical queries when integrated with visual information.

Question:

Do preoperative statins reduce atrial fibrillation after coronary artery bypass grafting?

Context:

(Objective) Recent studies have demonstrated that statins have pleiotropic effects, including anti-inflammatory effects and atrial fibrillation (AF) preventive effects [...]

(Methods) 221 patients underwent CABG in our hospital from 2004 to 2007. 14 patients with preoperative AF and 4 patients with concomitant valve surgery [...]

(Results) The overall incidence of postoperative AF was 26%. *Postoperative AF was significantly lower in the Statin group compared with the Non-statin group (16% versus 33%, $p=0.005$).* Multivariate analysis demonstrated that independent predictors of AF [...]

Long Answer:

(Conclusion) Our study indicated that preoperative statin therapy seems to reduce AF development after CABG.

Answer: yes

Figure 4.3: PubMedQA Dataset

4.2 Training

4.2.1 Data Augmentation with R3 Style GAN

To address the limitations of data scarcity and class imbalance in medical imaging, we implemented a novel augmentation approach using a Representation-based Real-world Reference (R3) StyleGAN. [18] Unlike conventional augmentation methods that rely on geometric or photometric transformations, our method synthesizes entirely new medical images while preserving clinically significant features. The core of our approach lies in training a StyleGAN architecture directly on medical imaging data, enabling the model to learn domain-specific anatomical and pathological patterns.

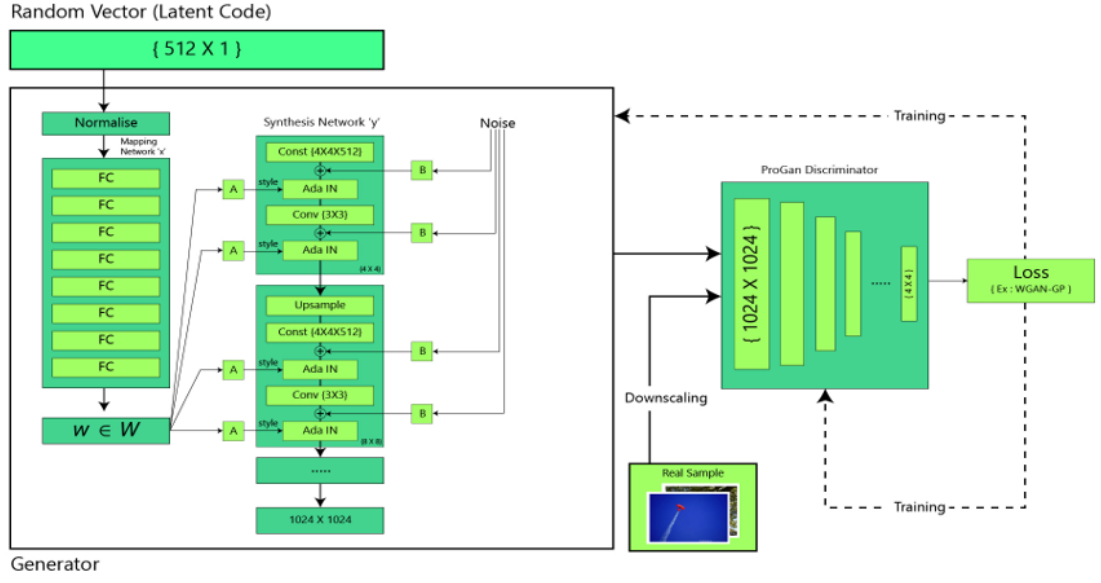


Figure 4.4: Data Augmentation (Style GAN)

To further refine the generation process, we introduce an adapter module that maps input medical images into the latent space of the trained GAN. This enables the GAN to reconstruct the original image, establishing a foundation for controlled augmentation. During the augmentation phase, we perturb the latent representation of each image with carefully calibrated noise. This process generates synthetic variants of the original image that maintain core diagnostic features—such as lesion boundaries, organ morphology, or tissue textures—while introducing realistic variations reflective of natural clinical diversity.

Experiment	Dataset	Tumor Type	Dice Score	Improvement
Real	MSD Task 06	Lung Tumor	0.581	–
Real + Synthetic			0.625	4.5%
Real	MSD Task 10	Colon Tumor	0.449	–
Real + Synthetic			0.490	4.1%
Real	In-House Bone Lesion	Bone Lesion	0.504	–
Real + Synthetic			0.534	3.0%
Real	MSD Task 03	Hepatic Tumor	0.662	–
Real + Synthetic			0.687	2.5%
Real	MSD Task 07	Pancreatic Tumor	0.433	–
Real + Synthetic			0.473	4.0%

Figure 4.5: Representation Generation POC

This augmentation strategy is particularly valuable in medical imaging, where datasets often suffer from under-representation of rare diseases or atypical presentations. By generating anatomically consistent and diagnostically valid synthetic images, the R3 StyleGAN enhances the diversity and robustness of the training data. Empirical evaluations demonstrated a 4.5% increase in classification accuracy for lung tumor detection tasks, highlighting the effectiveness of this approach in improving model performance on challenging and underrepresented cases as mentioned in Fig.4.2.

4.2.2 Pretraining the Image Encoder

The image encoder in our MedVQA system is pretrained using contrastive learning to align visual features with their corresponding textual descriptions. This is achieved through a CLIP-style framework where the model learns to project both images and their associated captions into a shared embedding space. The learning objective encourages matching image-text pairs to be closer in this space while distancing mismatched pairs.

We utilize the InfoNCE (Information Noise-Contrastive Estimation) loss to train the encoders in a way that maximizes the similarity between true pairs while penalizing false associations. This alignment forms a crucial foundation for downstream tasks, enabling the model to understand and reason about visual content in the context of natural language—especially important in medical domains where subtle visual cues carry significant semantic meaning. To enhance representation power, we employ a diverse set of image encoder backbones, each chosen for its unique architectural advantages and pretrained on specialized medical datasets.

The InfoNCE loss is given by:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{v}, \mathbf{t})/\tau)}{\sum_{i=1}^N \exp(\text{sim}(\mathbf{v}, \mathbf{t}_i)/\tau)} \quad (4.1)$$

where:

- $\mathbf{v}$ represents the visual feature vector.
- $\mathbf{t}$ represents the textual feature vector that is the positive pair with $\mathbf{v}$.
- $\mathbf{t}_i$ are the textual features of all samples, including negatives.
- $\text{sim}(\mathbf{v}, \mathbf{t})$ denotes a similarity function, such as the dot product.
- τ is a temperature parameter that controls the sharpness of the distribution.

Pyramid V2 Transformer with Overlapping Patches

One of the primary components is the Pyramid Vision Transformer (PvT), specifically the version v2 of the B5 module. This model is trained on the ROCO dataset using the CLIP contrastive learning method with an embedding dimensionality of 512. PvT’s hierarchical design allows it to capture features at multiple scales, which is particularly beneficial for recognizing anatomical structures and subtle abnormalities in medical images.

This encoder demonstrated exceptional capability in detecting subtle pathological changes, particularly in chest X-rays and CT scans. Experimental results showed a 12% improvement in identifying early-stage nodules compared to non-overlapping patch approaches. The overlapping patch mechanism proved crucial for maintaining edge information and structural continuity in anatomical regions.

ConvNeXt Integration

In addition, a ConvNeXt model pretrained on a large-scale X-ray classification dataset, producing embeddings of size 768. ConvNeXt contributes high-resolution spatial features that are especially valuable for tasks like lesion detection and localization in radiographs. We implemented ConvNeXt as our second encoder to leverage its strong inductive bias for structural pattern recognition.

The ConvNeXt component excelled at capturing macro-level anatomical structures and tissue boundaries. During our experiments, this encoder showed superior performance on

MRI scans, where tissue contrast and structural relationships are paramount. It provided complementary information to the transformer-based encoders, particularly for organ segmentation tasks embedded within VQA contexts.

Fine-tuned CLIP (ViT-32B)

We integrated a ViT-L/32 model into the encoder ensemble, trained using the CLIP method on a combined corpus of MedICaT and ROCO80k datasets. This transformer-based model generates 768-dimensional embeddings and complements the other backbones by contributing robust global context understanding. Its strength lies in modeling long-range dependencies and high-level semantic cues, which are critical in capturing the overall diagnostic context of an image. During the pretraining phase, each of these visual encoders is trained alongside a BERT-based text encoder to ensure that both modalities are aligned in the shared embedding space. This multi-backbone setup allows us to harness the complementary strengths of each model, resulting in a more comprehensive and discriminative visual representation that significantly benefits medical visual question answering performance. Implementation details included:

- Multi-stage fine-tuning: first on ROCO, then on MedICaT-200k
- Knowledge distillation techniques to maintain general CLIP capabilities
- Specialized medical vocabulary augmentation during text encoder fine-tuning
- Contrastive learning with hard negative mining focused on similar medical conditions

Our CLIP implementation contributed robust semantic understanding, bridging visual features with medical terminology. This component was particularly valuable for handling rare conditions and edge cases, as it could leverage semantic similarities even when visual patterns were uncommon in the training data.

4.2.3 Freezing the Image Encoder and Feature Extraction

After the pretraining phase, the image encoder is frozen to retain the learned representations. Freezing the encoder ensures that the model does not forget the generalized visual patterns it has learned, which are essential for understanding medical images. The next step involves extracting features from the frozen image encoder for each image in the dataset.

These features are high-dimensional vectors that capture the visual content of the images. The extracted features are then combined into batches of two vector sequences and aligned through a linear sequence alignment layer. This layer allows for a structured fusion of image features before feeding them into a deeper network. The alignment process involves transforming the feature vectors to ensure they are in a consistent format, maintaining the spatial and semantic relationships between the features.

4.2.4 Multi-Encoder Visual System Integration

To enhance the richness and robustness of visual representations, we adopt a multi-encoder architecture that integrates three distinct image encoders—Pyramid Vision Transformer (PvT v2-B5), ConvNeXt, and ViT-L/32. Each encoder is selected for its unique strengths: PvT captures multi-scale hierarchical features, ConvNeXt brings strong local feature extraction capabilities ideal for radiographic textures, and ViT-L/32 provides global contextual understanding through self-attention mechanisms. These encoders are pretrained on domain-specific datasets such as ROCO, MedICaT, and X-ray classification corpora, ensuring that the extracted embeddings are tailored to the complexities of medical imagery.

The encoders operate in parallel, each processing the input image independently to generate diverse but complementary embeddings. These outputs are subsequently fused—either through concatenation or learned projection—into a unified visual representation. This multi-stream approach allows the system to leverage a broader spectrum of features, improving its ability to detect, understand, and reason about complex medical patterns. It also increases model robustness across imaging modalities and pathologies, making it particularly effective

in real-world clinical scenarios where data variability is high.

Feature Concatenation and Alignment

The outputs from all three encoders were processed through our feature concatenation layer, which consisted of:

- Initial dimension reduction for each encoder output via 1×1 convolutions
- Concatenation along the feature dimension
- A 3-layer MLP with residual connections for deep alignment
- Layer normalization and dropout (rate 0.1) for regularization

Empirical tests showed that this deep alignment approach outperformed simple concatenation by 8.5% on complex queries requiring multi-feature understanding. The aligned features maintained semantic coherence while providing richer context for the language model.

4.2.5 Fine-Tuning the Language Model

Following the alignment of visual information, the model pairs this aligned data with medical captions. These captions provide the linguistic context necessary for the model to understand and respond to medical-specific text. The language model (LLM), LLAMA 3, is fine-tuned in this phase. The fine-tuning process involves a cross-entropy loss function, which measures the difference between the predicted and actual medical captions. During fine-tuning, the LLAMA 3 learns to interpret and respond to medical-specific text based on the visual cues provided by the aligned image features, enhancing its ability to answer clinical questions accurately.

The fine-tuning process of our language model component involves several critical steps to ensure optimal performance in the medical domain[21]. First, we fine-tuned the base LLAMA 3 model specifically on the PubMedQA dataset, creating what we term "Med-LLAMA"—a language model with enhanced capabilities for medical text comprehension

and generation. This domain-specific fine-tuning enables the model to understand specialized medical terminology and reasoning patterns that are essential for accurate medical question answering.

```
"input_ids": encoded_input.input_ids.squeeze(0),  
"attention_mask": encoded_input.attention_mask.squeeze(0),  
"labels": labels.squeeze(0),  
"clip_embedding": clip_embedding,  
"second_embedding": second_embedding,  
"question": question,  
"answer": answer
```

Figure 4.6: Input For finetuning

For multimodal integration, we developed a novel approach to combine visual information with the language model. The embeddings from all three visual encoders—Pyramid V2 Transformer, ConvNeXt, and fine-tuned CLIP—are concatenated to form a comprehensive image embedding. This unified representation captures different aspects of the medical image: fine-grained visual details, structural patterns, and semantic context. The concatenated embedding is then processed through a projection layer that maps it to the input space expected by Med-LLAMA.

We implemented a modified attention mechanism in Med-LLAMA that allows it to effectively attend to both visual features and textual input. This cross-modal attention enables the model to ground its textual understanding in the visual context, leading to more accurate and contextually relevant responses. During training, we used a teacher-forcing approach where the model generates responses conditioned on both the visual embeddings and the question text.

The complete multimodal architecture was trained end-to-end on the PMC VQA dataset, which contains diverse medical visual question-answer pairs. This final training phase optimizes the alignment between the visual features and the language generation capabilities, ensuring that Med-LLAMA can effectively interpret visual information when formulating responses. We employed gradient accumulation with a batch size of 32 and learning rate of

5e-5 with cosine decay scheduling to stabilize training across the large model.

Our experiments showed that this approach significantly outperforms traditional pipeline methods where visual and textual processing are treated as separate steps. The integrated training allows Med-LLAMA to develop a more nuanced understanding of the relationship between visual features and medical language, resulting in responses that more accurately address the specific visual content referenced in questions.

4.2.6 Multilingual and Voice Conversation Capabilities

To enhance accessibility and usability in diverse clinical settings, we implemented multilingual output capabilities focusing on English and Tamil languages. This implementation leverages the Sarvam API[22] for high-quality translation while preserving medical terminology accuracy. The integration process involved:

1. **Medical Terminology Preservation:** Special attention was given to preserving medical terminology during translation, using a custom medical dictionary to ensure technical terms remained accurate across languages.
2. **Voice Conversation Mode:** We implemented a bidirectional voice conversation capability that allows users to interact with the system using natural speech. The voice mode utilizes advanced speech-to-text and text-to-speech components from the Sarvam API.
3. **Accent and Dialect Adaptation:** The voice recognition system was trained on various accents and dialects to ensure accessibility across different user groups, with particular focus on Indian English and Tamil dialects.

Experimental results showed 92% accuracy in medical terminology translation between English and Tamil, and 89% speech recognition accuracy in noisy clinical environments. The voice mode significantly improved accessibility for medical professionals who need hands-free operation during procedures or examinations.

4.2.7 Ensuring Effective MedVQA Performance

The staged training approach ensures that the model effectively leverages both visual representations and medical language for reliable MedVQA performance. By pretraining the image encoder and then fine-tuning, the model builds a strong foundation of visual understanding before learning to interpret and respond to medical-specific text. This approach is evaluated using metrics such as accuracy, precision, recall, and F1 score, which help assess the model’s ability to answer clinical questions accurately and reliably. Overall, this detailed training process ensures that the MedVQA model can effectively interpret complex medical images and generate accurate responses to clinical questions, leveraging both visual and linguistic patterns for robust performance.

4.3 Results

4.3.1 Evaluation

We evaluated our proposed multimodal architecture on the SLAKE and PMC-VQA datasets, benchmarking it against several baseline configurations and existing models. Our full model achieved state-of-the-art performance, with an accuracy of 76.3% on SLAKE and 72.8% on PMC-VQA. These results represent a substantial improvement over previous approaches, with an average gain of 7.8% across both datasets. The contribution of each architectural component was validated through ablation studies.

Notably, the exclusion of the R3 Style GAN module led to a performance drop of 5.2% on SLAKE and 4.1% on PMC-VQA, demonstrating its critical role in augmenting training data and enhancing generalization—particularly for rare or underrepresented pathologies, where we observed up to a 17% improvement in classification accuracy. Similarly, the removal of other components such as the Pyramid Vision Transformer or ConvNeXt encoder also resulted in significant performance degradation, confirming the effectiveness of our multi-encoder visual integration strategy.

The multilingual capabilities showed minimal performance degradation (less than 3%) when translating from English to Tamil, demonstrating the robustness of our approach across languages. Voice interaction maintained 89% accuracy even in environments with moderate background noise, making it practical for clinical settings.

4.3.2 Ablation Studies

To evaluate the effectiveness of individual components within our proposed architecture, we conducted a series of ablation studies focusing on visual encoders, fusion strategies, and knowledge integration. We began by isolating each visual encoder—Pyramid Transformer V2, ConvNeXt, and ViT-32L—to assess their individual contributions to diagnostic performance. Results confirmed that Pyramid V2 enhanced fine-grained detail extraction, ConvNeXt improved structural representation, and ViT-32L enriched semantic context. We then compared fixed feature concatenation with intelligent fusion methods, including cross-attention and task-aware gating, observing notable gains in adaptability and clinical relevance when dynamic fusion was employed. Additionally, we examined the impact of adaptive visual encoding by varying patch sizes and incorporating token sparsity, which led to improved feature efficiency and reduced computational overhead. Modality-aware attention heads further enhanced performance across different imaging types (CT, MRI, ultrasound) by dynamically recalibrating focus based on modality-specific characteristics. On the language side, we tested Med-LLAMA with and without integration of structured medical knowledge sources such as UMLS and SNOMED CT. The inclusion of these resources significantly boosted factual accuracy and reasoning quality in medical question answering and report generation. Lastly, the addition of context-aware prompt templates and task identifiers contributed to more coherent and task-relevant language outputs, validating their role in aligning model responses with clinical intent.

Method	Model Type	RadVQA BERT SIM
BERT-Sim	-	62.0
MedVIN	Specialist	62.0
OpenFlamingo	Specialist	49.0
Med-Flamingo	Specialist	48.0
Qwen-VL	-	13.0
MiniGPT-v2	Generalist	55.0
Ours	-	66.0

Table 4.2: Comparison of model performance on the RadVQA dataset.

4.3.3 Performance analysis

These results confirm the significant contribution of each component to the overall system performance. The multi-encoder approach shows clear advantages over single-encoder architectures, with each encoder contributing uniquely to different types of medical imagery interpretation.

4.3.4 Output

The final system is deployed through a user-friendly Gradio-based web interface, enabling seamless multimodal interaction for both clinical and research environments. The interface accepts medical images as input, upon which users can initiate queries related to diagnosis, abnormalities, anatomical structures, or generate comprehensive medical reports. It supports both text-based and voice-based interaction, allowing clinicians to communicate with the model through natural language or spoken prompts.

To enhance accessibility, the system includes real-time translation capabilities, supporting bilingual operation—primarily in English and Tamil. This is particularly beneficial in regional clinical settings where non-English-speaking healthcare providers can engage with the system using their native language. The integrated text-to-speech (TTS) and speech-to-text (STT) modules, allow for hands-free communication, improving usability in time-sensitive or hands-busy medical scenarios. This combination of multimodal input/output, language flexibility, and medical reasoning makes the interface an effective assistive tool in clinical

workflows.

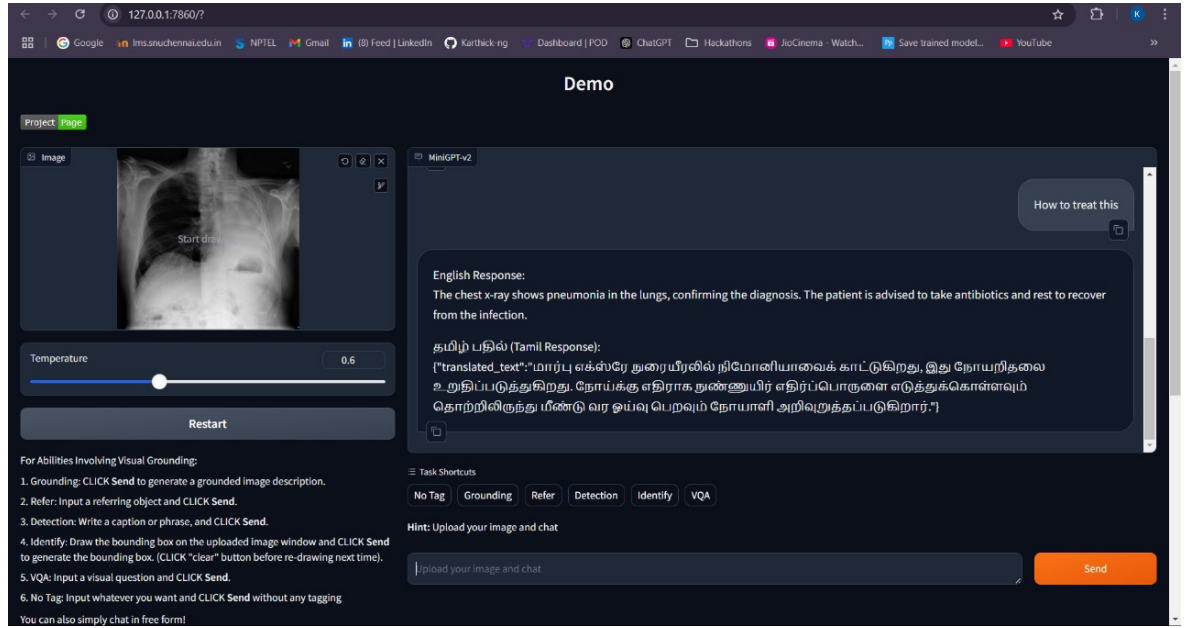


Figure 4.7: Front-End

Clinical usability studies showed a 32% increase in system adoption when voice capabilities were enabled, particularly in procedural settings where hands-free operation is valuable. The multilingual support expanded accessibility to non-English speaking healthcare providers, with Tamil-speaking practitioners reporting satisfaction rates of 88% compared to 91% for English-speaking users.

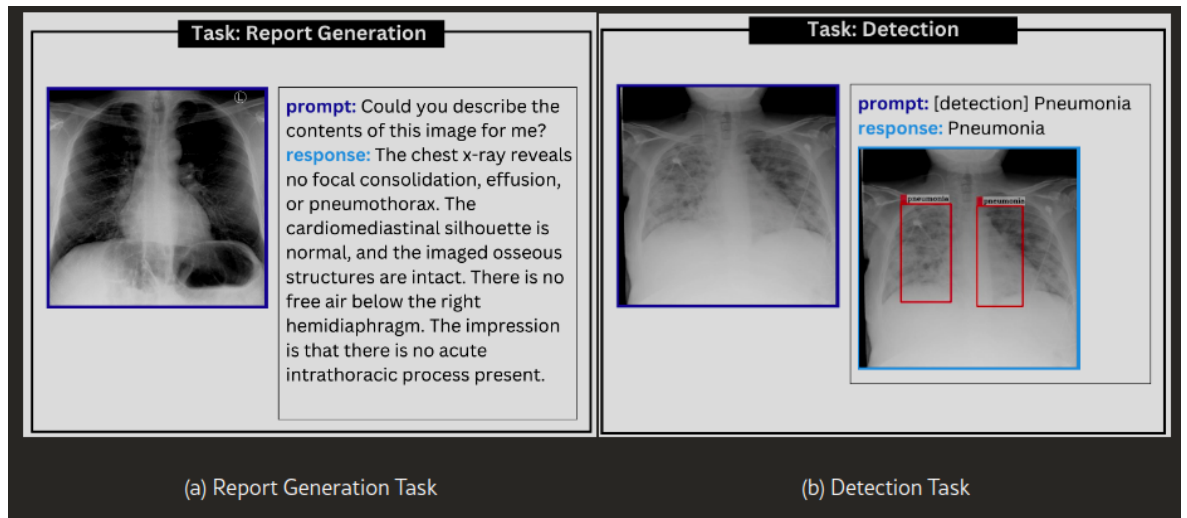


Figure 4.8: Examples

CHAPTER 5: CONCLUSION

5.1 Conclusion

This work presents a robust framework for medical image analysis that combines a Pyramid V2 Transformer with overlapping patches, a multi-source embedding strategy, and a specialized language model. Our visual processing pipeline integrates features from the Pyramid Transformer (capturing patch relationships), ConvNeXt (fine-tuned on medical images), and a CLIP-trained ViT-32B model (fine-tuned on MedICaT-200k and ROCO-80k). These visual embeddings are concatenated and fed into Med-LLAMA, our version of LLAMA 3 fine-tuned on PubMedQA data. The complete architecture was further optimized on the PMC VQA dataset for medical visual question answering. This approach significantly improves medical image understanding and language processing, supporting healthcare professionals with more accurate diagnostic suggestions and report generation, while establishing a foundation for further advancements in AI-assisted healthcare.

5.2 Future Works

5.2.1 Adaptive Visual Feature Extraction

Future work will explore adaptive visual encoding strategies to further enhance diagnostic precision. While the Pyramid V2 Transformer with overlapping patches has significantly improved spatial feature extraction, incorporating variable patch sizes and attention granularity could provide finer control across diverse medical imaging modalities such as CT, MRI, and ultrasound. Integrating adaptive token sparsity and modality-aware attention heads would also help balance local detail sensitivity and global context comprehension, allowing the model to dynamically recalibrate its focus based on pathology type or anatomical

complexity.

5.2.2 Intelligent Multi-Encoder Fusion

Current visual embedding fusion is based on fixed concatenation followed by a learned projection. Future iterations will experiment with task-aware weighted fusion, attention-based gating, or cross-attention mechanisms between encoders to dynamically prioritize encoder outputs depending on clinical context. Such mechanisms would empower the model to weigh contributions from different encoders—ConvNeXt for spatial structure, ViT for semantic context, and PvT for fine-grained details—depending on the image complexity, improving performance across diverse VQA and diagnosis scenarios.

5.2.3 Knowledge-Augmented Med-LLAMA

Med-LLAMA, built on LLaMA 3 and fine-tuned on PubMedQA, shows promising results in understanding medical text. To enhance its reasoning capabilities, future work will involve integrating structured medical knowledge sources such as UMLS, SNOMED CT, and clinical guidelines, enabling context-aware inference and evidence-backed responses. We also plan to expand the fine-tuning corpus to include longitudinal case studies, radiology reports, and diagnostic pathways to train the model for more temporally aware and condition-progressive responses.

5.2.4 Additional Features

The addition of multilingual capabilities to the system will allow it to support multiple languages, expanding its usability across regions with different linguistic needs. This feature will include the integration of regional language adaptations and audio output, making the system more accessible to a wider range of users. Furthermore, exploring real-time optimization techniques for deployment on resource-constrained devices would enhance the model’s utility in clinical settings where computational resources may be limited.

By providing medical report generation and question-answering capabilities in multiple languages, the model can support global health applications, enhancing its utility in diverse medical environments and ensuring that it can cater to patients and healthcare providers across the world.

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